

Supplementary Information for “Towards Interpretable Drug-Drug Interaction Prediction: A Graph-Based Approach with Molecular and Network-Level Explanations”

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A Related Work

Graph-based methods have been widely adopted for drug-drug interaction (DDI) prediction due to their ability to model complex and multi-faceted relationships. These methods are typically divided into two categories: homogeneous and heterogeneous graphs. Unlike homogeneous graphs, such as DDI networks that focus solely on drug nodes, heterogeneous graphs incorporate multiple biological entities to provide a more comprehensive representation of biological systems, making them particularly valuable for DDI prediction. This section reviews recent advancements in graph-based methods, which have significantly improved the accuracy of DDI prediction.

A.1 Molecular Graph-Based Methods

Molecular graphs represent the internal structure of drugs at the atomic level, with nodes as atoms and edges as chemical bonds, providing essential molecular insights for DDI prediction. Graph neural networks (GNNs) are commonly used to learn topological and semantic information from molecular graphs, resulting in drug-level representations for predictive tasks [4, 5]. However, such approaches treat the molecular graph as a whole, overlooking the influence of localized, chemically meaningful substructures that play a pivotal role in modeling DDIs. To address this, attention mechanisms have been introduced to evaluate pairwise interactions between substructures identified through node neighborhoods [9, 13] or adaptive and learnable modules [12, 21, 22]. While these mechanisms have improved prediction accuracy, they often yield substructures that are either incomplete or lack clear chemical semantics. Recently, Zhong et al. [25] utilized a chemistry-inspired molecular fragmentation function to decompose drugs into chemically meaningful substructures, enabling more precise interaction

modeling and improved predictive performance. Although these methods are well-suited to handle cold-start scenarios and scale easily to novel drugs, their reliance on sparse DDI data and limited integration of broader biological context constrain their overall effectiveness.

A.2 Biomedical Knowledge Graph-Based Methods

Biomedical knowledge graphs (BKGs) integrate diverse biological entities—such as drugs, proteins, and diseases—as nodes, and their various interactions (e.g., drug-protein, protein-disease) as edges, offering a macroscopic view of biological interactions essential for DDI prediction. Early studies, such as HIN-DDI [17], relied on hand-crafted meta-paths to capture the topological features of drug pairs within BKGs, while Karim et al. [7] utilized node embedding techniques to extract drug representations, which were later fed into machine learning or deep learning models for prediction. The emergence of GNN-based methods has significantly improved the predictive power of BKGs by learning expressive node embeddings that aggregate topological and feature-based information [26]. KGNN [10], for instance, pioneered sampling subgraphs from BKGs to extract local neighborhood features, capturing the semantic and topological properties of drug nodes. Subsequent methods [6, 15, 20, 23] improved on this by combining drug-drug interaction networks with external BKGs to construct closed subgraphs for drug pairs, employing diverse modeling techniques to extract pair-specific semantics and topologies for more accurate DDI prediction. Despite these advancements, these methods often struggle to provide high-quality representations for drugs with weakly connected neighbors, leading to decreased performance, particularly in scenarios involving incomplete data.

A.3 Multi-level Integration Methods

Researchers have explored methods that integrate graph-level and node-level representations to achieve a comprehensive understanding of drug interactions. Graph-level representations are typically derived from molecular graphs, capturing the internal structural features of drugs, while node-level features are obtained from interaction networks centered around drugs, such as DDI networks [19] or BKGs [11, 16], which provide contextual information about

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a drug’s biological interactions. Current approaches usually concatenate or merge these single-drug features to produce unified embeddings, which are then combined with another drug’s embedding for pairwise interaction prediction. However, this strategy treats drug pairs as separate entities, failing to model their interactions as a unified entity, potentially struggling to capture the intricate and interdependent nature of drug interactions fully, and limiting predictive accuracy.

B Experimental Settings

B.1 Datasets

DDI Datasets. In this study, we utilized two publicly available benchmark datasets for multi-class DDI prediction: Ryu’s dataset [14] and an updated version of the DrugBank dataset (version 5.1.12, released March 14, 2024) [8]. Both datasets are derived from the DrugBank database, a comprehensive repository of biochemical and pharmaceutical information focusing on DDIs at the metabolic level, such as those that reduce serum concentrations or increase bleeding risks and severity. Ryu’s dataset, developed by Ryu et al., originates from an earlier version of DrugBank and comprises 1,709 drugs, forming 191,388 drug pairs annotated with 86 interaction types. Most drug pairs (99.8%) have a single interaction type, with only a tiny fraction exhibiting multiple types. To ensure consistency and support robust 5-fold cross-validation, Ryu et al. excluded edges with more than one interaction type and maintained at least five drug pairs for each interaction category. Following a similar methodology, we processed the more recent 2024 DrugBank database to construct an expanded and updated dataset. This newer version comprises 2,152 drugs forming 617,277 edges annotated with 221 interaction types, representing a significant increase in scale and detail compared to the earlier dataset.

external Knowledge Graph. To support DDI prediction tasks, we constructed a drug-centered biomedical knowledge graph, DrugCentricBioKG (Fig. 1), by extending the existing biological knowledge graph BioKG [18] with additional data sourced from Bioteque [3]. DrugCentricBioKG comprises seven primary entity types, including drugs, proteins, and diseases, and captures their associations through 12 distinct relationship types, such as drug-target and drug-disease interactions. The graph comprises 241,838 nodes and 2,308,645 edges, offering a large-scale, semantically rich resource tailored for drug-centric biomedical applications. Detailed statistics on the entity and relationship types are presented in Table 1.

B.2 Baselines

We evaluate the performance of our proposed MolecBioNet framework against several widely recognized and state-of-the-art methods for DDI prediction:

- **DeepDDI** [14]: A deep learning-based model that predicts DDIs by utilizing drug pair similarity profiles derived from their chemical structures.
- **SSI-DDI** [13]: A GAT-based approach that constructs drug representations by capturing neighboring substructures, using a co-attention mechanism to infer substructure interactions for DDI prediction.

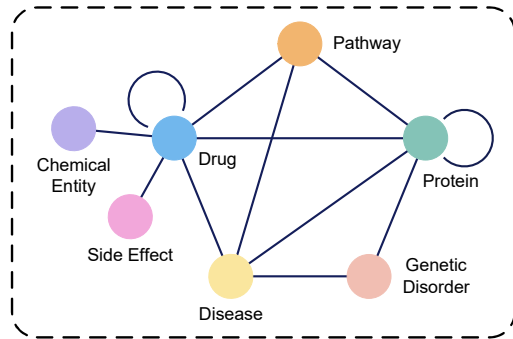


Figure 1: Metagraph of the DrugCentricBioKG, showing all the entities and associations between them.

- **GGI-DDI** [22]: A granular computing-based model that identifies critical substructures within molecular graphs and models their interactions to predict DDIs precisely.
- **KGNN** [10]: An innovative approach that samples subgraphs from BKGs to capture local neighborhood features, which are then used to generate effective drug embeddings for DDI prediction.
- **SumGNN** [23]: A self-attention-driven model that synthesizes key reasoning paths within subgraphs, employing multi-channel integration to utilize external biomedical knowledge and enhance performance on complex multi-typed DDI tasks.
- **LaGAT** [6]: A link-aware graph attention model that generates adaptive attention pathways based on specific drug pair links, enabling better capture of semantic relationships for DDI prediction.
- **KnowDDI** [20]: A GNN-based approach that enhances drug representations by adaptively aggregating rich neighborhood information from extensive biomedical knowledge graphs and learning interpretable subgraphs for each drug pair.
- **TIGER** [16]: A dual-channel relation-aware graph transformer model utilizes the combined representation learning from drug molecular graphs and biomedical knowledge graphs to enhance DDI prediction performance.

B.3 Setup for Novel Drug Evaluation Tasks

To evaluate the performance of MolecBioNet on novel drugs, we design two experimental tasks: (1) Novel Drug–Existing Drug (ND-ED) experiments, which predict interactions between novel drugs and existing drugs, and (2) Novel Drug–Novel Drug (ND-ND) experiments, which predict interactions between pairs of novel drugs.

In this setting, the complete set of drugs denoted as $\mathcal{D}$, is divided into three mutually exclusive subsets: $\mathcal{D} = \mathcal{D}_{train} \cup \mathcal{D}_{valid} \cup \mathcal{D}_{test}$, where $\mathcal{D}_{train} \cap \mathcal{D}_{valid} \cap \mathcal{D}_{test} = \emptyset$. Here, $\mathcal{D}_{train}$ contains existing drugs used for training, $\mathcal{D}_{valid}$ consists of novel drugs for validation, and $\mathcal{D}_{test}$ comprises novel drugs for testing.

For both tasks, the training set consists of samples $\mathcal{S}_{train} = \{(u, r, v) \mid u, v \in \mathcal{D}_{train}, r \in \mathcal{R}\}$. In the ND-ED experiments, the validation samples are defined as $\mathcal{S}_{valid} = \{(u, r, v) \mid u \in \mathcal{D}_{train}, v \in \mathcal{D}_{valid}, r \in \mathcal{R}\} \cup \{(u, r, v) \mid u \in \mathcal{D}_{valid}, v \in \mathcal{D}_{train}, r \in \mathcal{R}\}$, and the test samples as $\mathcal{S}_{test} = \{(u, r, v) \mid u \in \mathcal{D}_{train}, v \in$

Table 1: Entity and Relationship Counts in DrugCentricBioKG

Entity Type	Entity Count	Relationship Type	Relationship Count
Drug	178,415	Drug-Drug Interaction	1,176,303
Chemical Entity	26,658	Protein-Protein Interaction	395,174
Protein	18,607	Drug-Chemical Entity Link	303,810
Disease	5,962	Drug-Side Effect Association	136,781
Genetic Disorder	4,409	Drug-Disease Association	113,594
Side Effect	4,303	Protein-Pathway Association	95,556
Pathway	3,484	Protein-Disease Association	38,750
		Drug-Protein Interaction	34,333
		Drug-Pathway Association	5,064
		Disease-Genetic Disorder	4,621
		Protein-Genetic Disorder	4,578
		Disease-Pathway Association	81

$\mathcal{D}_{test}, r \in \mathcal{R}\} \cup \{(u, r, v) \mid u \in \mathcal{D}_{test}, v \in \mathcal{D}_{train}, r \in \mathcal{R}\}$. For the ND-ND experiments, the validation samples are defined as $S_{valid} = \{(u, r, v) \mid u, v \in \mathcal{D}_{valid}, r \in \mathcal{R}\}$, and the test samples as $S_{test} = \{(u, r, v) \mid u, v \in \mathcal{D}_{test}, r \in \mathcal{R}\}$.

B.4 Ablation Analysis

To investigate the contribution of each component within our MolecBioNet framework, we conduct ablation studies by designing four model variants, each with a specific feature or module removed:

- **w/o. tsBKG:** Removes the tsBKG-based subgraph representation for drug pairs, omitting the contextual information derived from biomedical knowledge graphs.
- **w/o. HIG:** Excludes the hierarchical interaction graph module, preventing the model’s ability to capture substructure interaction patterns.
- **w/o. Designed Pooling:** Replaces context-aware subgraph pooling (CASPool) and attention-guided influence pooling (AGIPool) with MeanPooling, ignoring relational importance and node influence, thereby assessing the contribution of pooling mechanisms.
- **w/o. InfoMin:** Eliminates the mutual information minimization loss term, aiming to assess its impact on reducing redundancy and improving representation at multiple levels.

B.5 Implementation

In our experiments, we adopt a 5-fold cross-validation strategy [2] to ensure robust evaluation. For each interaction type, we randomly split samples into the training set, validation set, and test set based on the ratio of 3:1:1. The optimization process leverages the Adam optimizer and fine-tune the hyperparameters, exploring learning rates in $\{10^{-2}, 10^{-3}, 10^{-4}\}$ and weight decay values in $\{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$. To effectively capture structural and contextual information from biomedical knowledge graphs, we extract

2-hop enclosing subgraphs for each drug pair, ensuring the inclusion of relevant relational patterns for DDI prediction. The training procedure spans 80 epochs, during which the auxiliary loss terms in the overall objective function are balanced by setting hyperparameters β and γ to 2 and 10, respectively. Further details about the data and code of MolecBioNet are available in our public GitHub repository at <https://github.com/MengjieChan/MolecBioNet>.

B.6 Evaluation Metrics

We assess the model performance using four widely accepted metrics for multi-class classification tasks. *Accuracy (ACC)* quantifies the proportion of correctly classified samples across the entire dataset, offering a global assessment of model performance. *Macro-averaged F1 Score (F1)* computes the F1 score independently for each class and then averages these values without weighting, providing an evaluation of performance in scenarios with imbalanced class distributions. *Precision-Recall Area Under the Curve (PR-AUC)* evaluates the trade-off between precision and recall across different decision thresholds, making it particularly suitable for datasets with significant label imbalances. Finally, *Cohen’s Kappa* measures the agreement between predictions and ground truth while accounting for random chance, serving as a robust indicator of model reliability. For all metrics, we report the mean and standard deviation of the test results obtained through 5-fold cross-validation, ensuring robust and reliable comparisons across models.

B.7 Interpretability Evaluation Metrics

We employ two quantitative metrics to evaluate the interpretability of model explanations: *Fidelity* [1] and *Sparsity* [24].

Fidelity measures the faithfulness of explanations by assessing how much the model’s prediction changes when the explanatory features are either removed (for necessity) or retained (for sufficiency). Specifically, *Fidelity₊* reflects the *necessity* of the explanation—higher values indicate that the removed features are essential to the model’s decision, whereas *Fidelity₋* reflects the *sufficiency* of

the explanation—lower values indicate that the retained subgraph alone is sufficient for correct prediction. These metrics are formally defined as:

$$Fidelity_+ = 1 - \frac{1}{N} \sum_{i=1}^N \mathbb{1}(\hat{y}_i^{\mathcal{G}_{C \setminus S}} = \hat{y}_i),$$

$$Fidelity_- = 1 - \frac{1}{N} \sum_{i=1}^N \mathbb{1}(\hat{y}_i^{\mathcal{G}_S} = \hat{y}_i),$$

where $\hat{y}_i$ denotes the original prediction for graph $\mathcal{G}_i = (\mathbf{A}, \mathbf{X})$. The subgraph $\mathcal{G}_S = (\mathbf{A} \odot \mathbf{M}_E, \mathbf{X} \odot \mathbf{M}_{VX})$ retains the important edges and node features as indicated by the masks $\mathbf{M}_E$ and $\mathbf{M}_{VX}$, which may be either hard (binary) or soft (continuous) representations of feature importance. The complement subgraph $\mathcal{G}_{C \setminus S}$ contains the unimportant components. Only node features are perturbed in our implementation, while the graph topology remains unchanged.

Sparsity measures the compactness of the explanation by quantifying the proportion of input features that are ignored. It is computed as:

$$Sparsity = \frac{1}{N} \sum_{i=1}^N \left(1 - \frac{|m_i|}{|M_i|}\right),$$

where $|m_i|$ denotes the number of features (nodes, edges, or node features) identified as important in explanation m_i , and $|M_i|$ represents the total number of features in graph $\mathcal{G}_i$. Higher sparsity values indicate more concise explanations focusing exclusively on the most informative components.

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